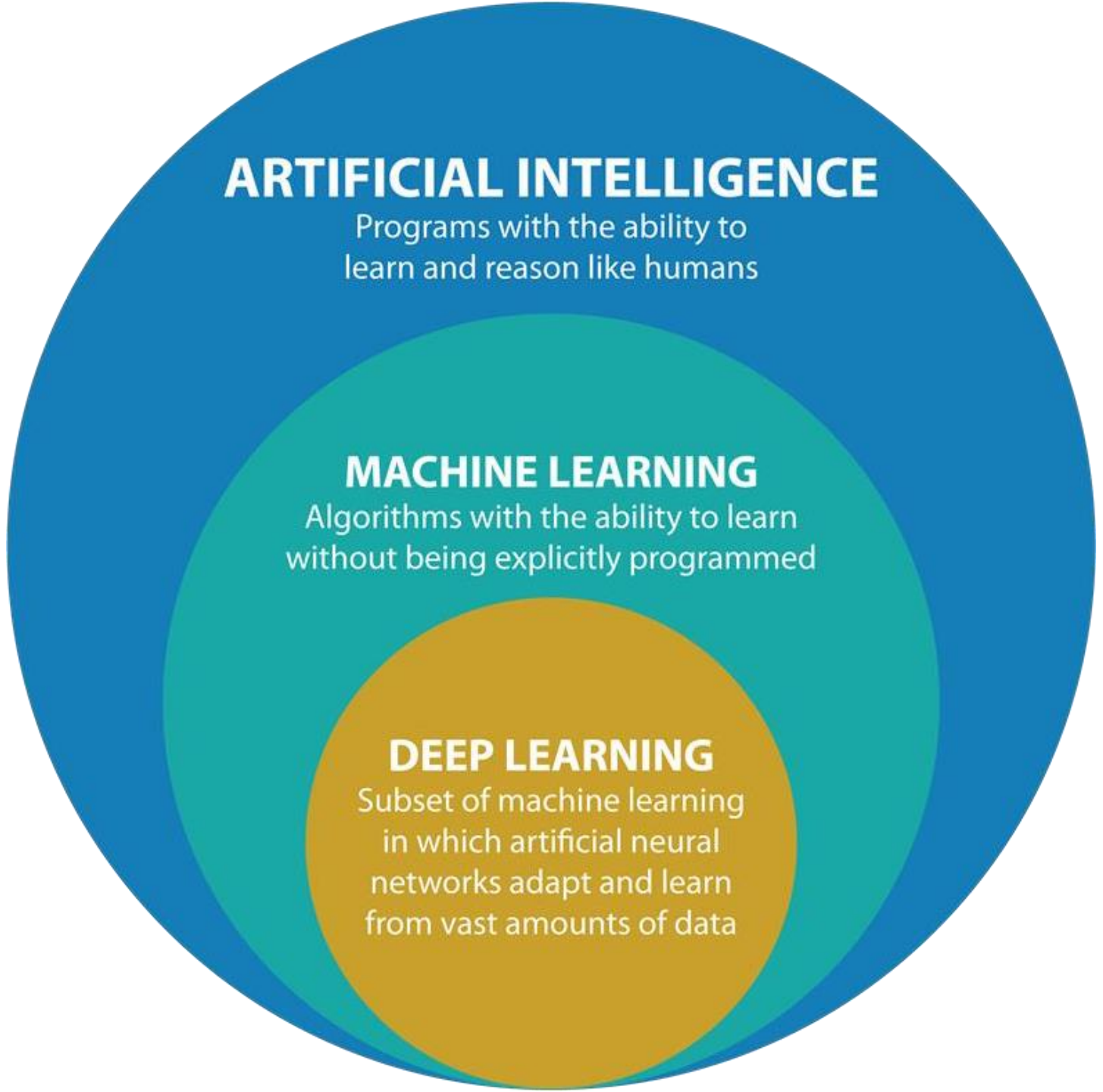
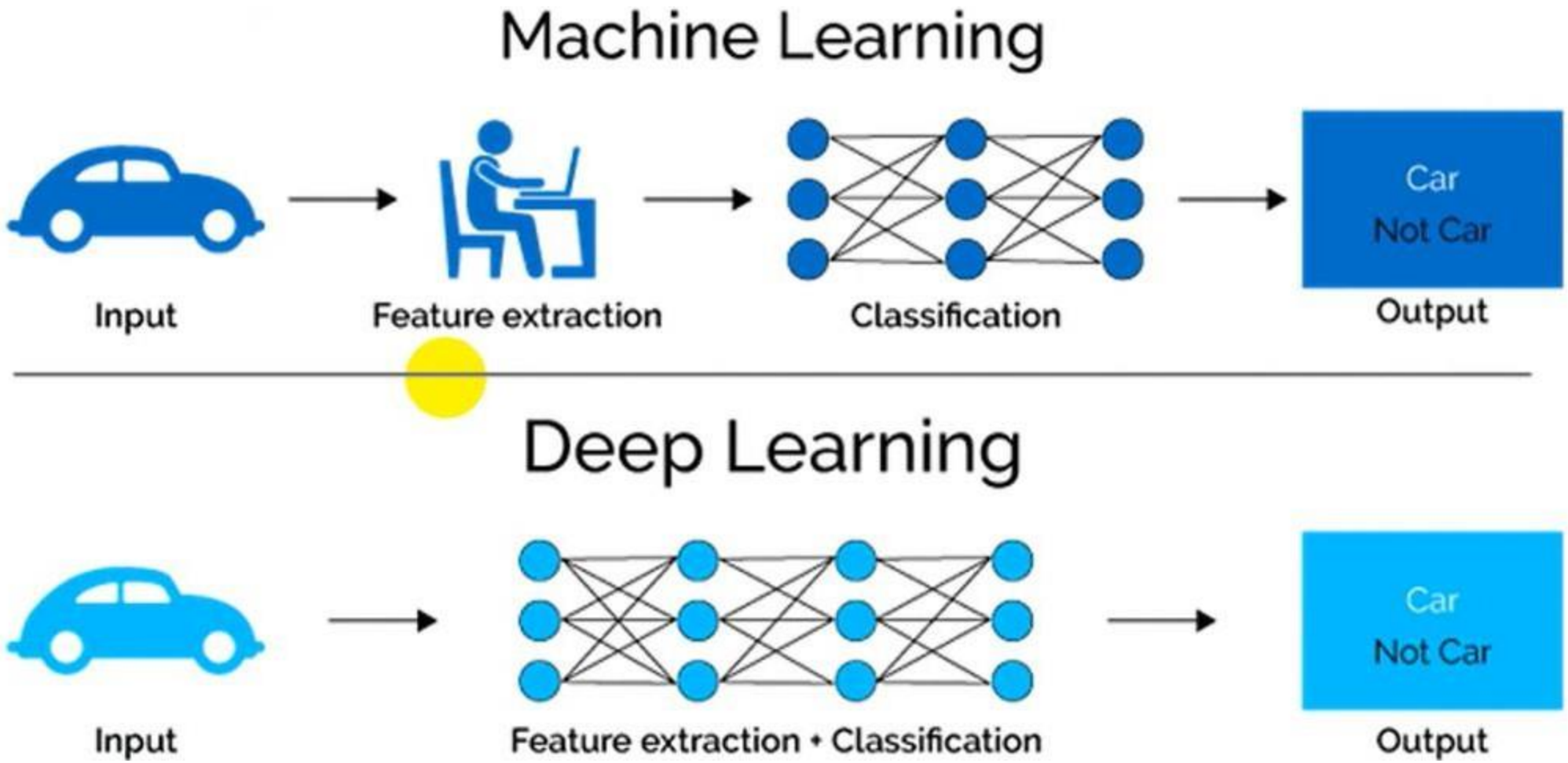

ARTIFICIAL INTELLIGENCE
Programs with the ability to learn and reason like humans
MACHINE LEARNING
Algorithms with the ability to learn without being explicitly programmed
DEEP LEARNING
Subset of machine learning in which artificial neural networks adapt and learn from vast amounts of data

AI Vs ML Vs DL

- **In AI we are mimicking human intellect and behavior**
- **In ML we are making machines to learning without explicitly being programmed**
- **In DL we are making machines to learning based on artificial neural networks**


Machine Learning
Input
Feature extraction
Classification
Car
Not Car
Output
Deep Learning
Input
Feature extraction • Classification
Car
Not Car
Output

Revising ML Concepts



Human can learn from past experience
and make decision of its own

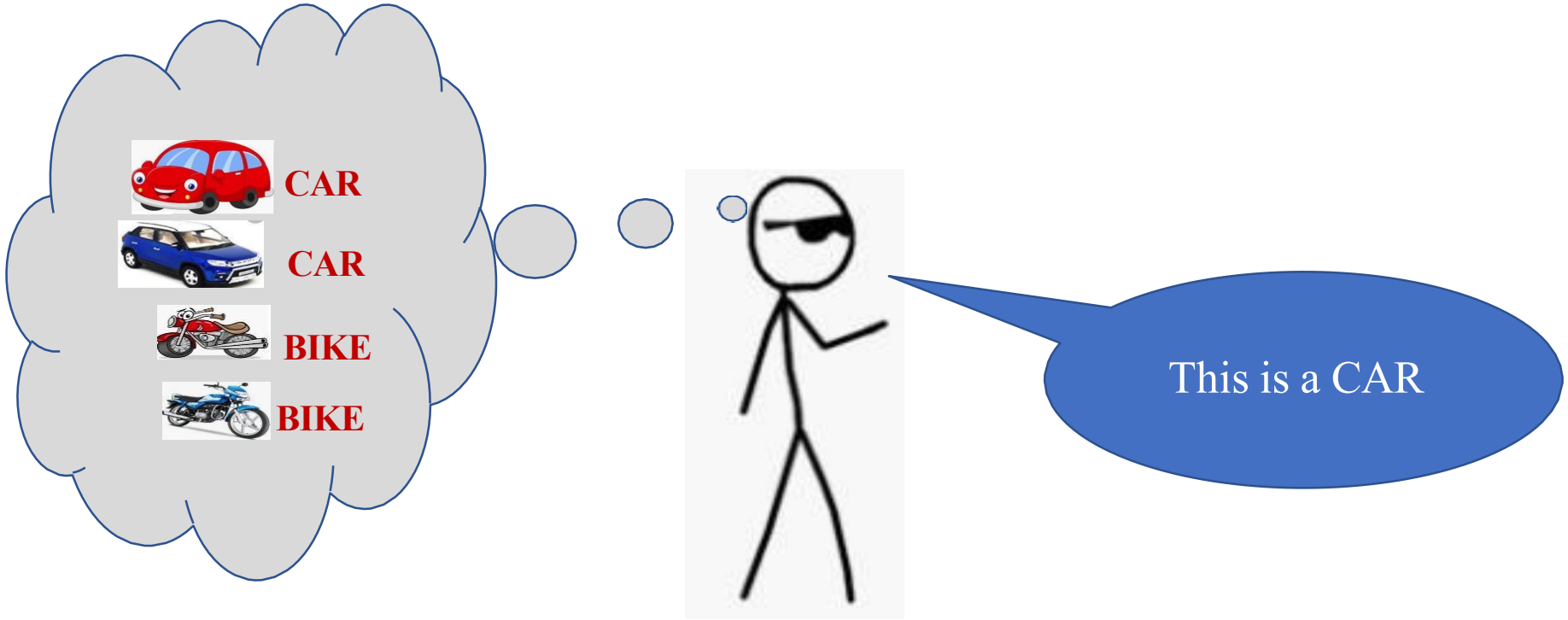


What is this object?





What is this?



Let us ask the same question to him

What is this?





Let us ask the same question to him



What is this?







[But, he is a human being. He can observe and learn]



show him











show him



CAR



CAR



BIKE



BIKE

What is this?





 **CAR**

 **CAR**

 **BIKE**

 **BIKE**

Past experience

Let us ask the same question now

CAR

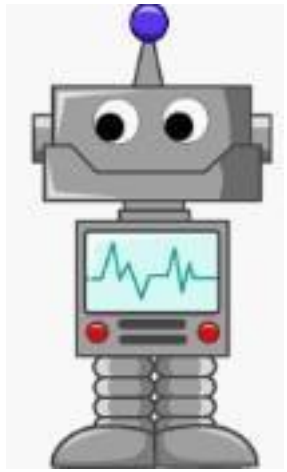
What is this?







What about a Machine ?



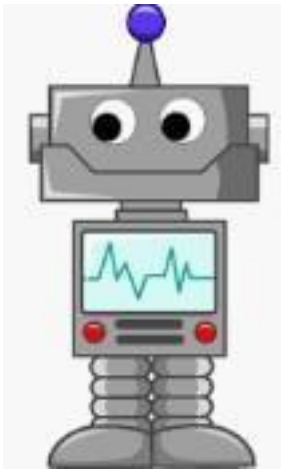
Machines follow instructions

It can not take decision of its own

What about a Machine ?

We can ask a machine

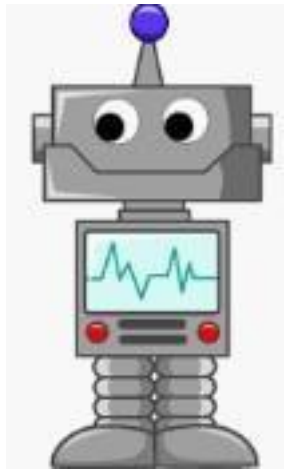
- To perform an arithmetic operations such as
  - Addition
  - Multiplication
  - Division



Machines follow instructions

What about a Machine ?

- Comparison
- Print



Machines follow instructions

What is Machine Learning?

We want a machine to act like a human

What is Machine Learning?





Identify this object

What is Machine Learning?





Price in 2025?

Predict the price

What is Machine Learning?

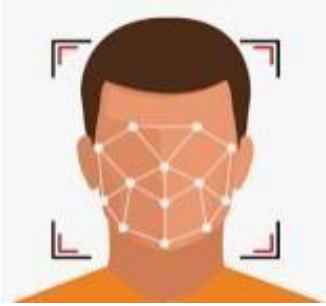


I ~~made~~ met him yesterday

Natural language understand, and correct grammar

What is Machine Learning?





recognize face

Recognize faces

What is Machine Learning?

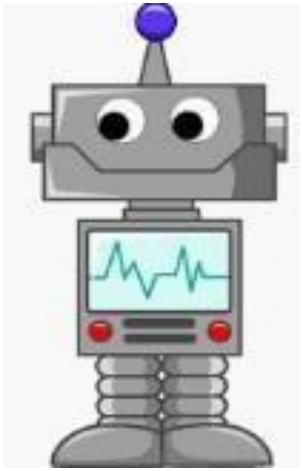


What do we do?

Just like, what we did to human, we need to

provide experience to the machine.

What is Machine Learning?



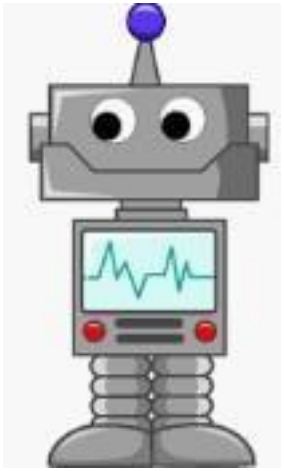
+



Dataset

First need to provide training dataset to the machine

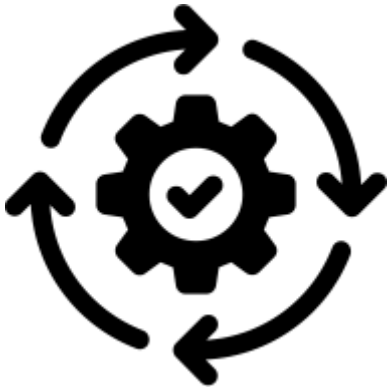
What is Machine Learning?



+

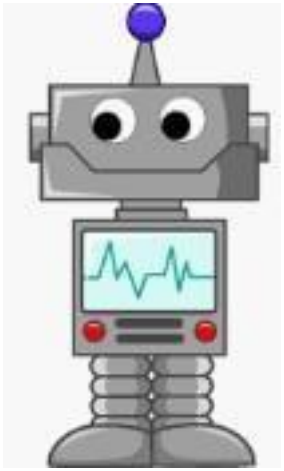


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Then, devise algorithms and execute programs on the data with respect to the underlying target tasks

What is Machine Learning?

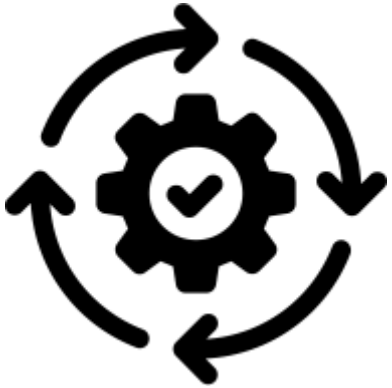


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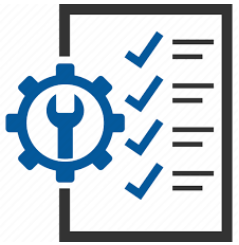


Dataset

+

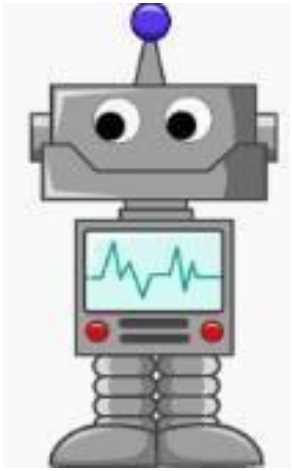


+



Then, using the programs, identify rules

What is Machine Learning?

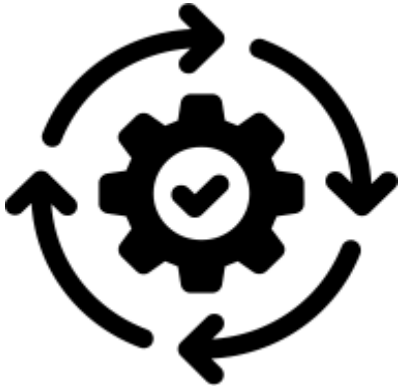


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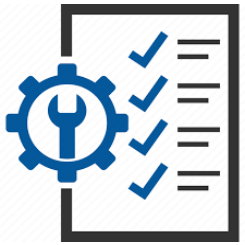
Dataset

+



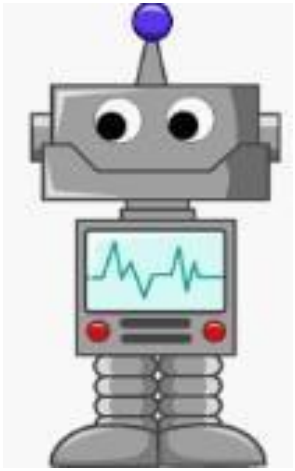
+





Extract required patterns

What is Machine Learning?

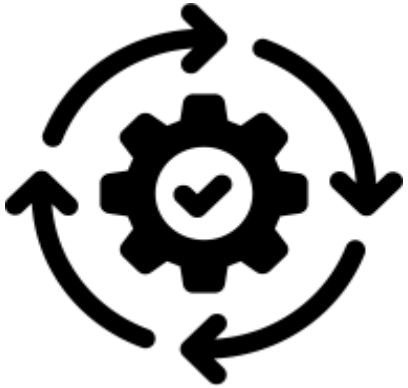


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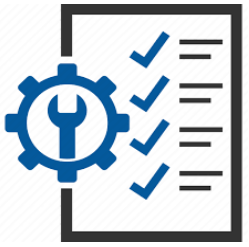


Dataset

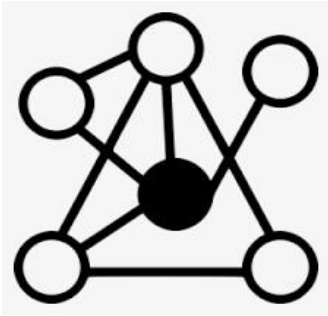
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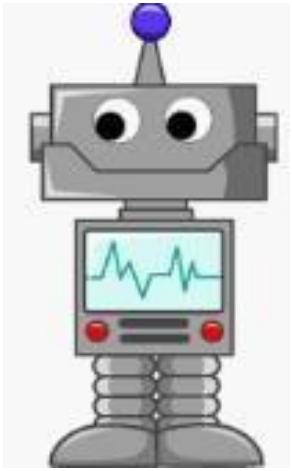






Identify relations

What is Machine Learning?

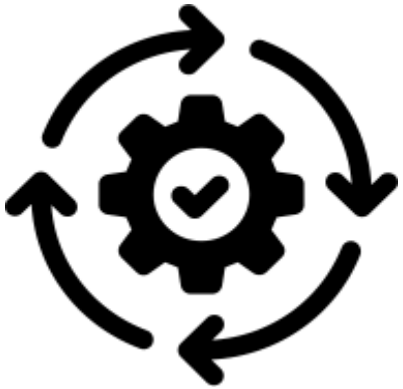


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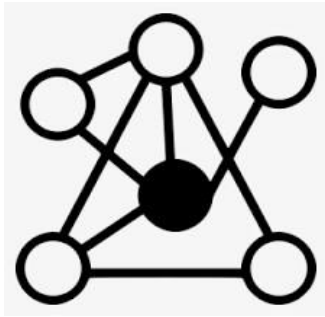
Dataset

+

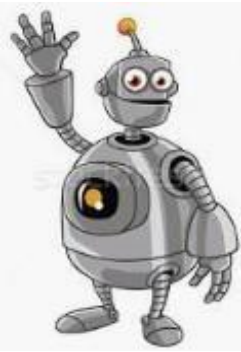


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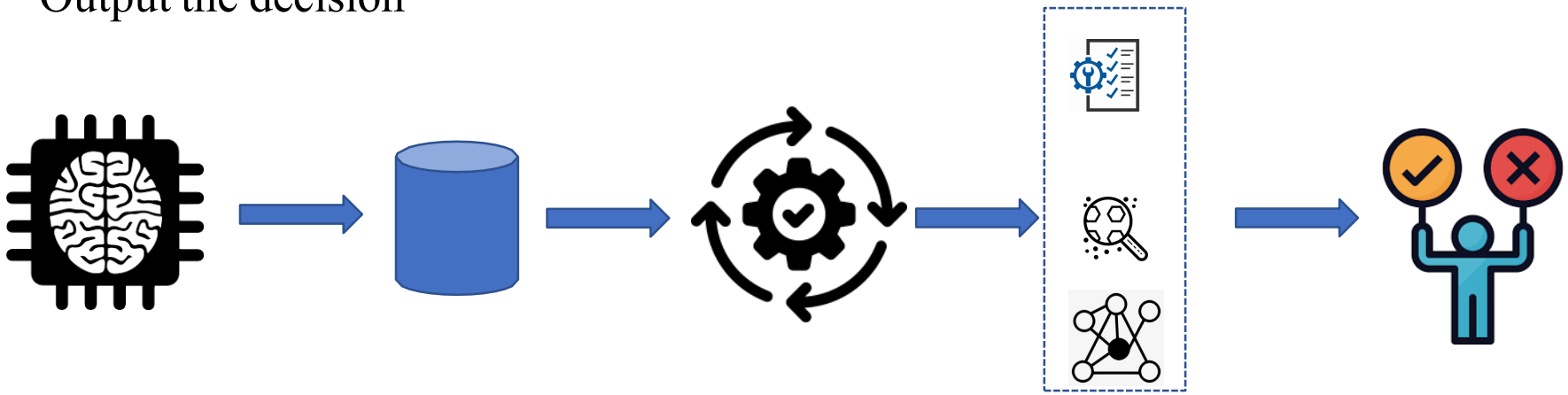


So that machine can derive inferences from the data

In summary, what is machine learning?

Given a machine learning problem

- Identify and create the appropriate dataset
- Perform computation to learn
  - Required rules, pattern and relations
- Output the decision

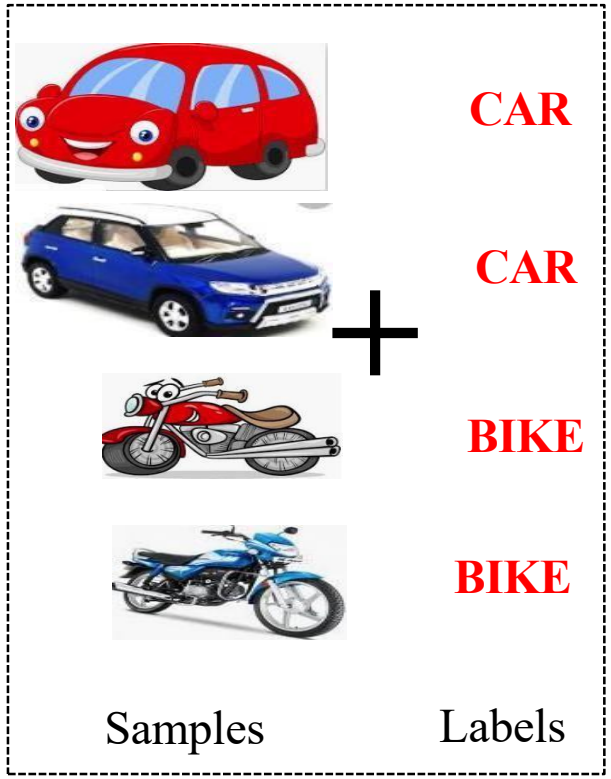


Machine Learning Paradigms

- Supervised
- Unsupervised Learning
- Reinforcement learning

We as human being solve various types of problem in our day-to-day life, Various decisions need to be taken.
Depending on the nature of the problem, machine learning tasks can be broadly divided in three categories.

What is Supervised Learning?



Training Dataset

In supervised learning, we need some thing called a Labelled Dataset



CAR



CAR

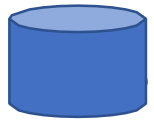


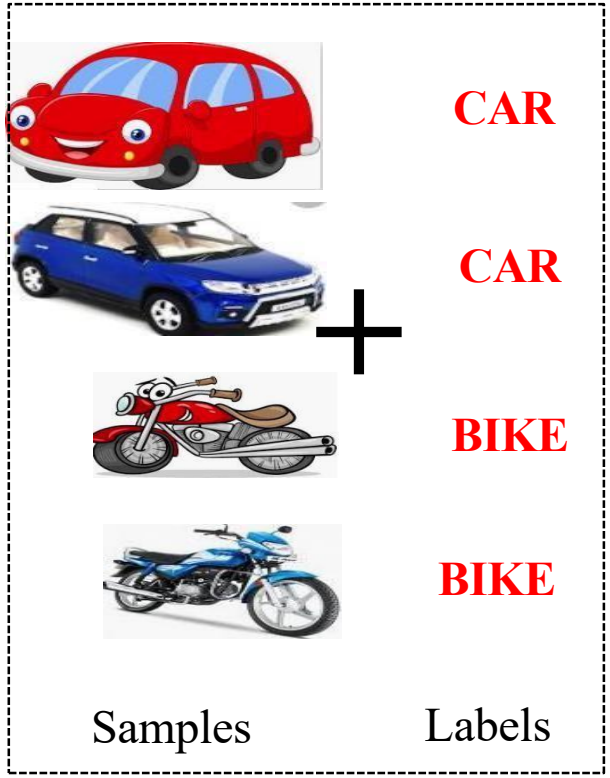
BIKE



BIKE

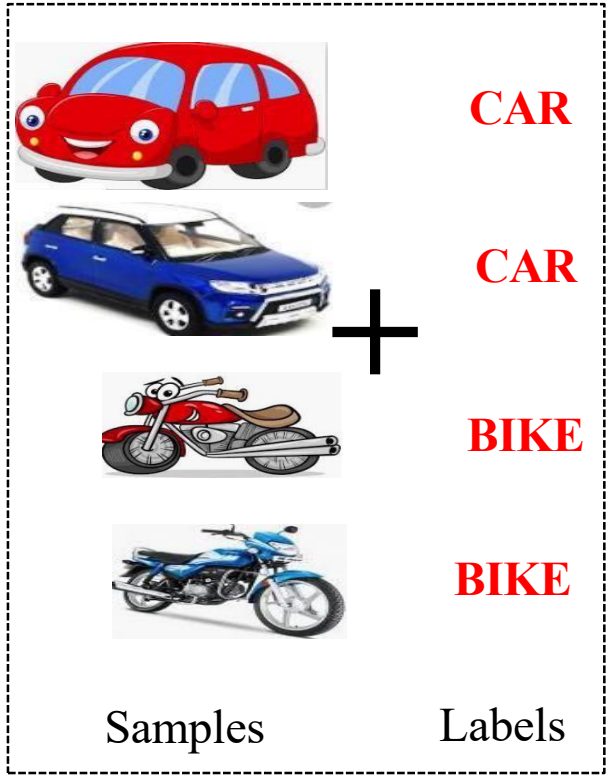
Samples Labels

+ = **Training Dataset** $f($  $)=$



= **Training Dataset**

$f($  $)=$



= **Training Dataset**

$f($  $= \text{CAR}$


CAR


CAR


BIKE


BIKE

Samples Labels

+ = Training Dataset

Classification

$f($  ,  $)=$ CAR

If the possible output values of the function are predefined and discrete/categorical, it is called classification.



CAR



CAR

+



BIKE



BIKE

Samples Labels

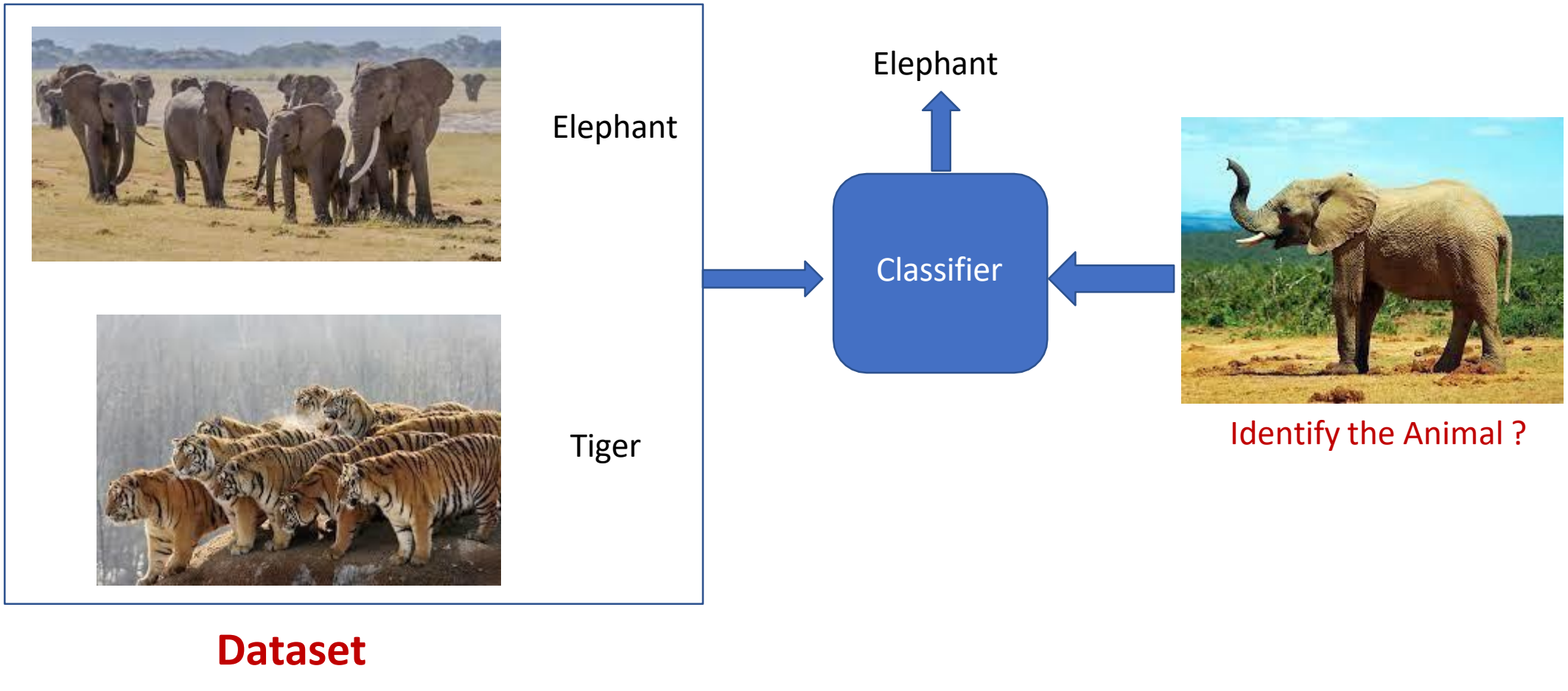
= Training Dataset

Classification

$f($  ,  $)=$ CAR

Predefined classes means, it will produce output only from the labels defined in the dataset. For example, even if we input a bus, it will produce either CAR or BIKE

Classifier


Elephant
Tiger
Dataset
Classifier
Elephant
Identify the Animal ?

Regression



Dataset

Regression

$f($  ,  $)$ = 20500.50

[If the possible output values of the function are continuous real values, then it is called Regression

The classification and Regression problems are supervised, because the decision depends on the characteristics of the ground truth labels or values present in the dataset, which we define as experience

What is Unsupervised Learning



~~CAR~~



~~CAR~~



BIKE



BIKE

Dataset

In unsupervised learning, we do not need to know the labels or Ground truth values .



Dataset



Clustering

The task is to identify the patterns like group the similar objects together

What is Unsupervised Learning









Dataset













Association Rules Mining

More Example Unsupervised Learning



Dataset

More Example Unsupervised Learning



Dataset



More Example Unsupervised Learning







What is Reinforcement Learning

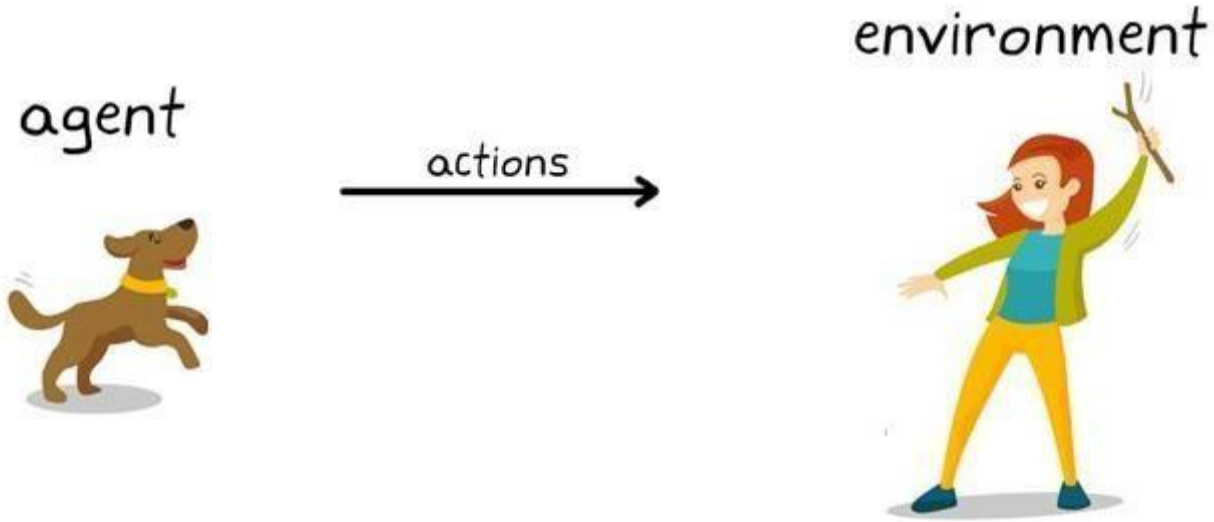
[It is also known as learning from trials and errors]

What is Reinforcement Learning

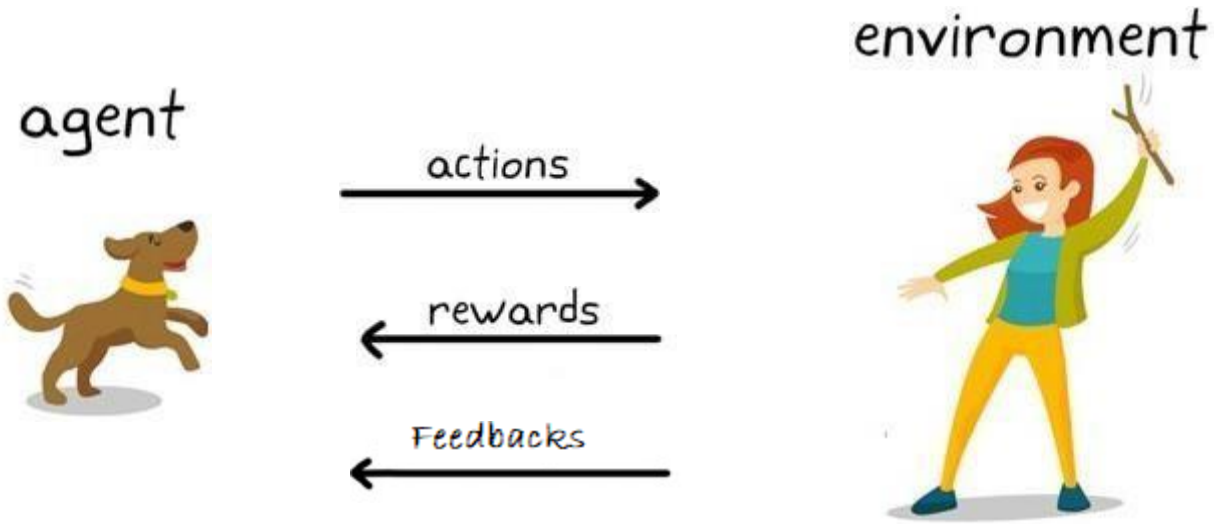




What is Reinforcement Learning



What is Reinforcement Learning



Another Example



Agent



Task



Environment

Reinforcement Learning









Punishment

Reinforcement Learning









Reward

Reinforcement Learning









Reward

Baby Learn from the Trials and Errors

Reinforcement Learning

Machine Learning

Input: X

Output: Y



Label”motorcycle”

Why is it hard?

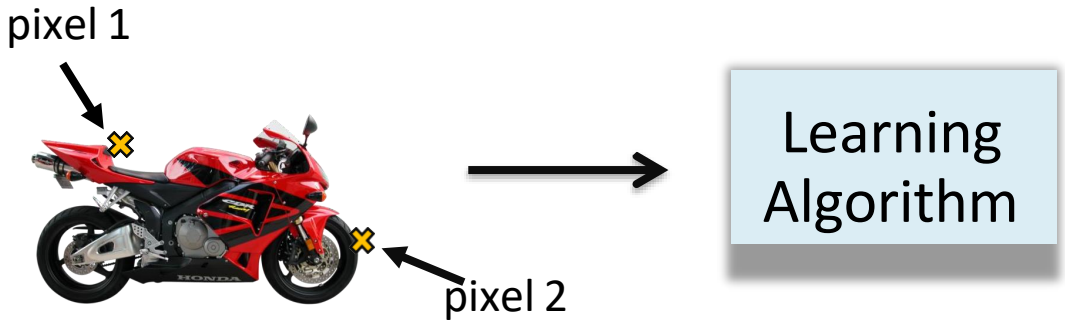
You see this

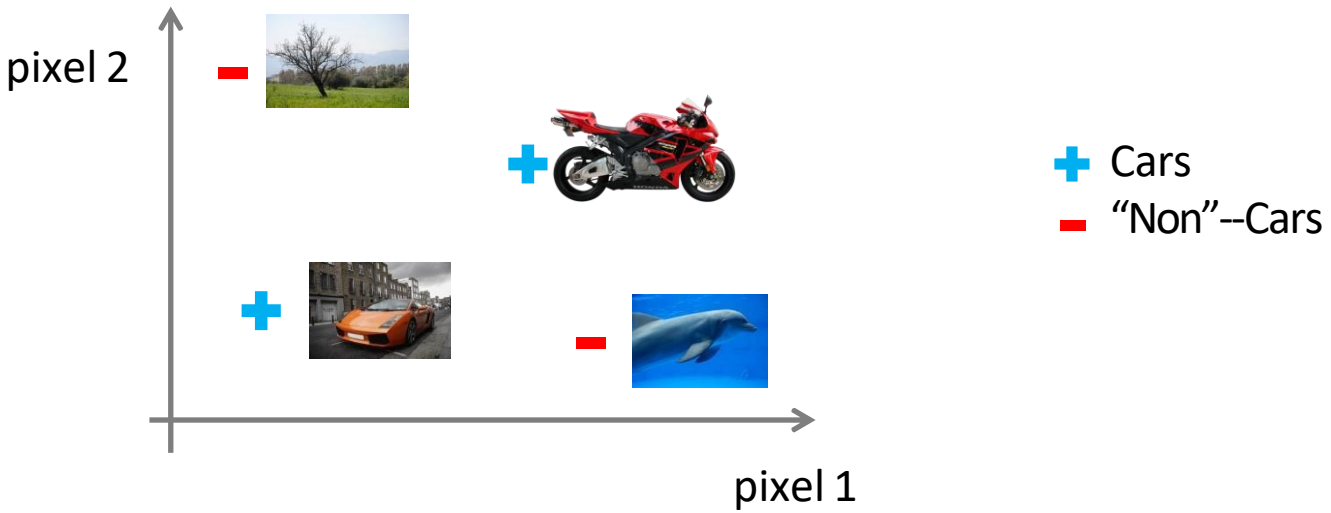


But the camera sees this:

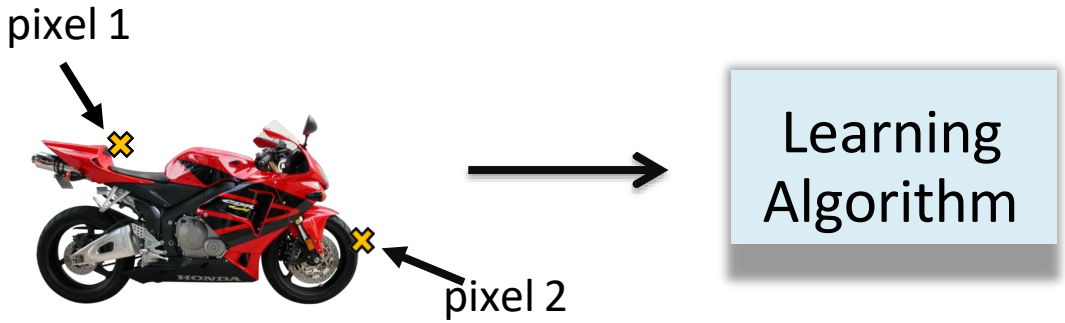
194	210	201	212	199	213	215	195	178	158	182	209
180	189	190	221	209	205	191	167	147	115	129	163
114	126	140	188	176	165	152	140	170	106	78	88
87	103	115	154	143	142	149	153	173	101	57	57
102	112	106	131	122	138	152	147	128	84	58	66
94	95	79	104	105	124	129	113	107	87	69	67
68	71	69	98	89	92	98	95	89	88	76	67
41	56	68	99	63	45	60	82	58	76	75	65
20	43	69	75	56	41	51	73	55	70	63	44
50	50	57	69	75	75	73	74	53	68	59	37
72	59	53	66	84	92	84	74	57	72	63	42
67	61	58	65	75	78	76	73	59	75	69	50

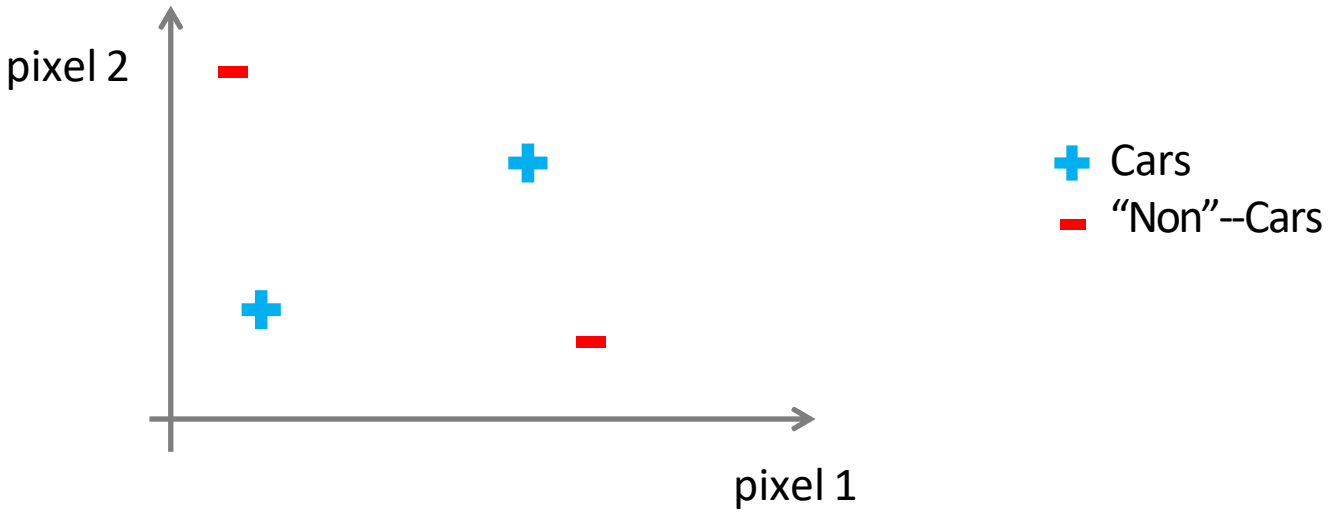
Raw Image Representation



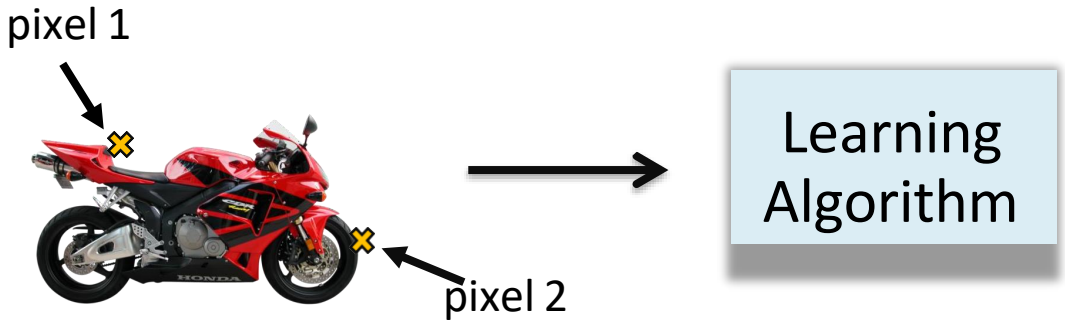


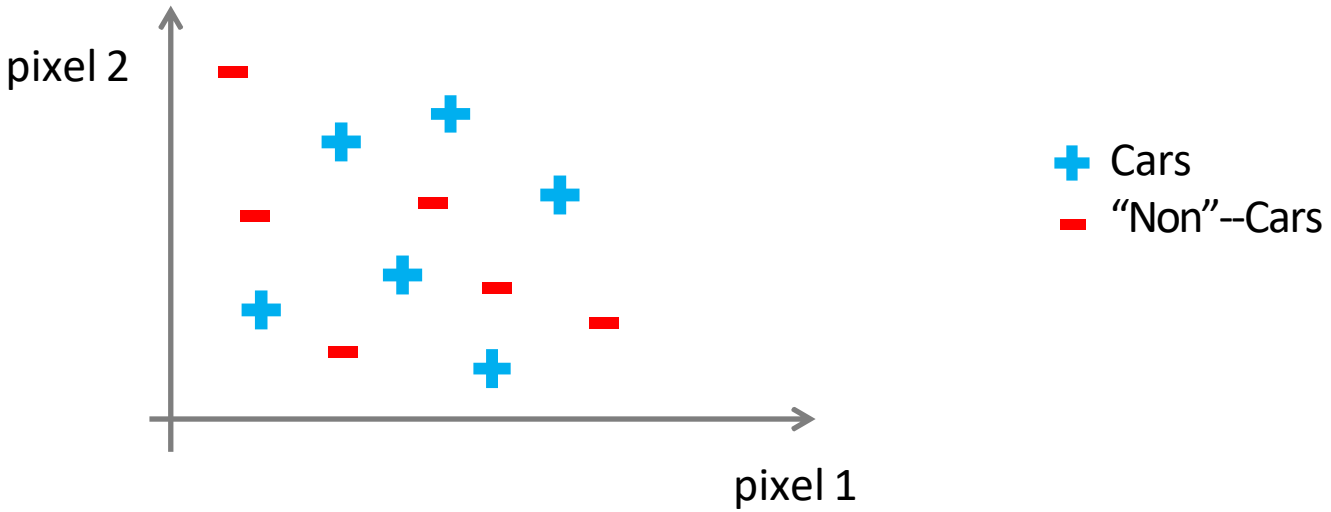
Raw Image Representation



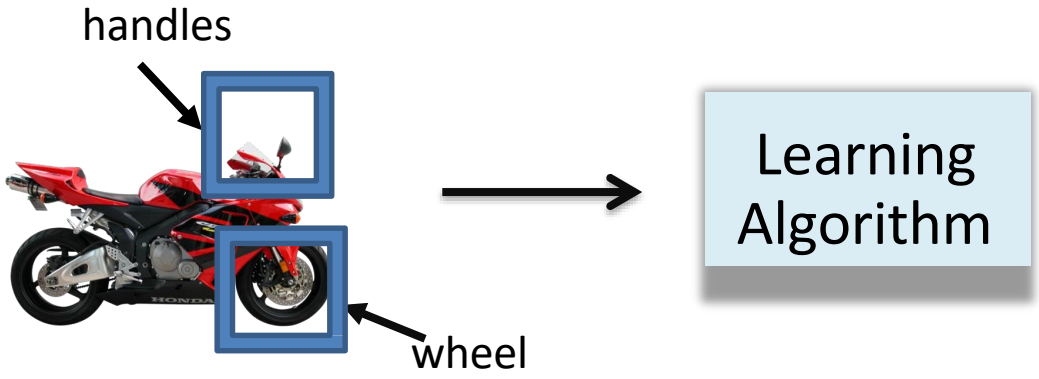


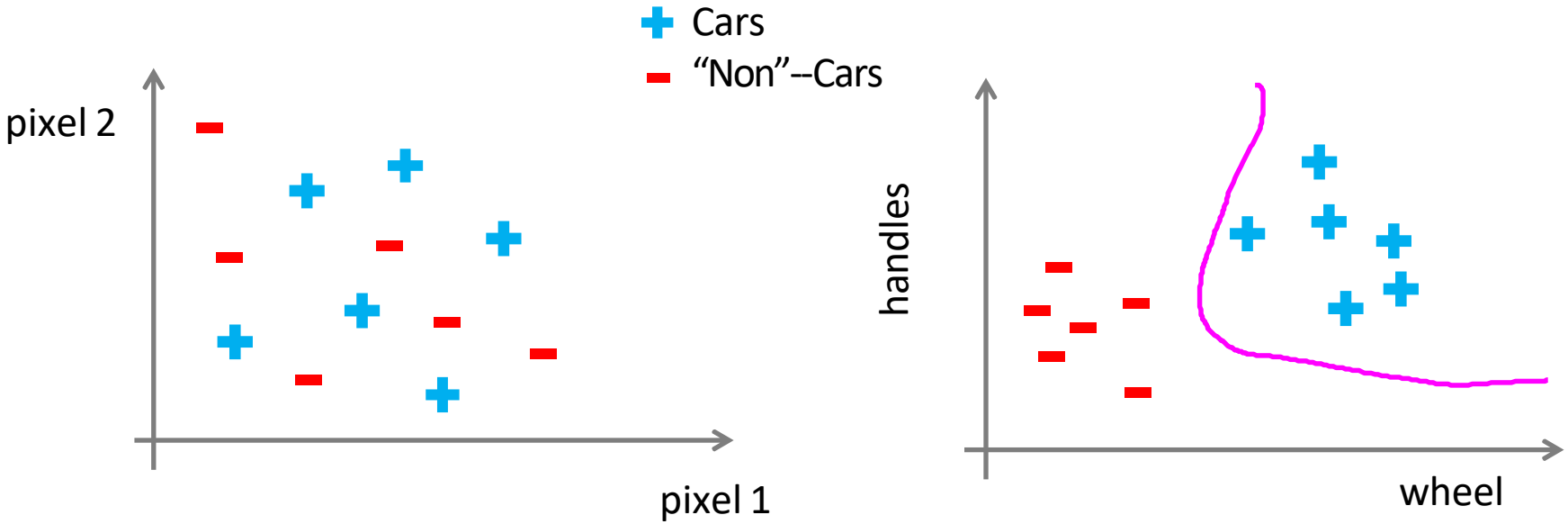
Raw Image Representation



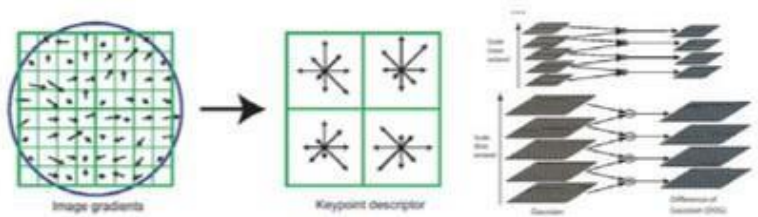


Better Feature Representation?

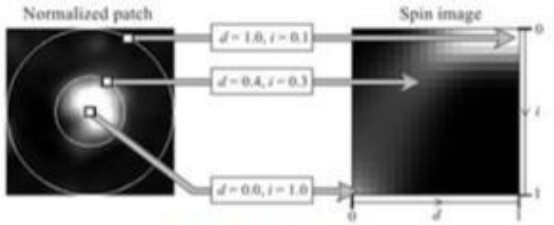




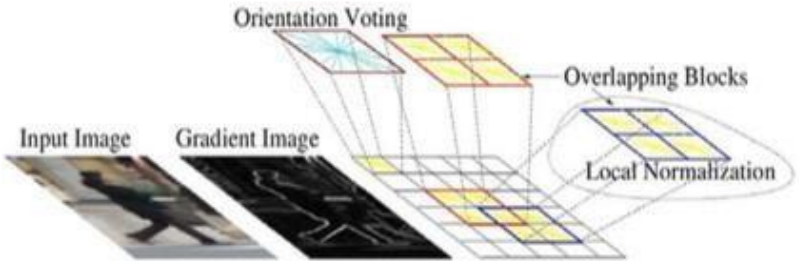
Feature Representations



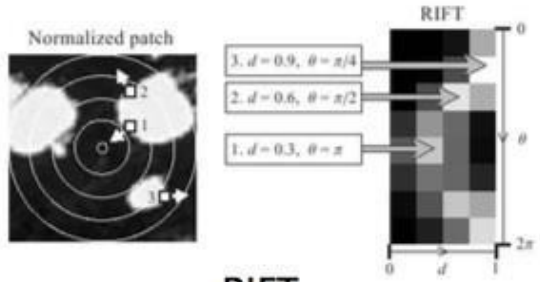
SIFT



Spin image

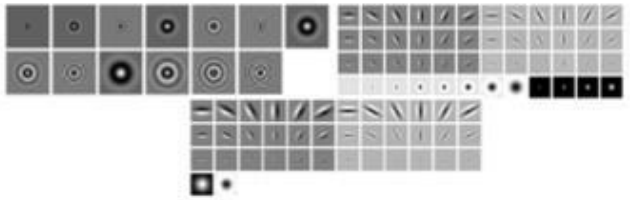


HoG

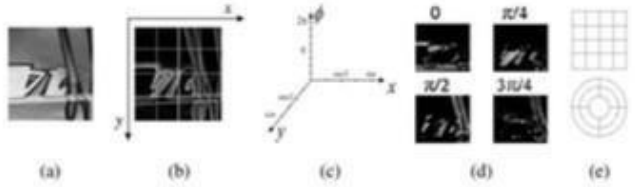


RIFT

Expert Knowledge!

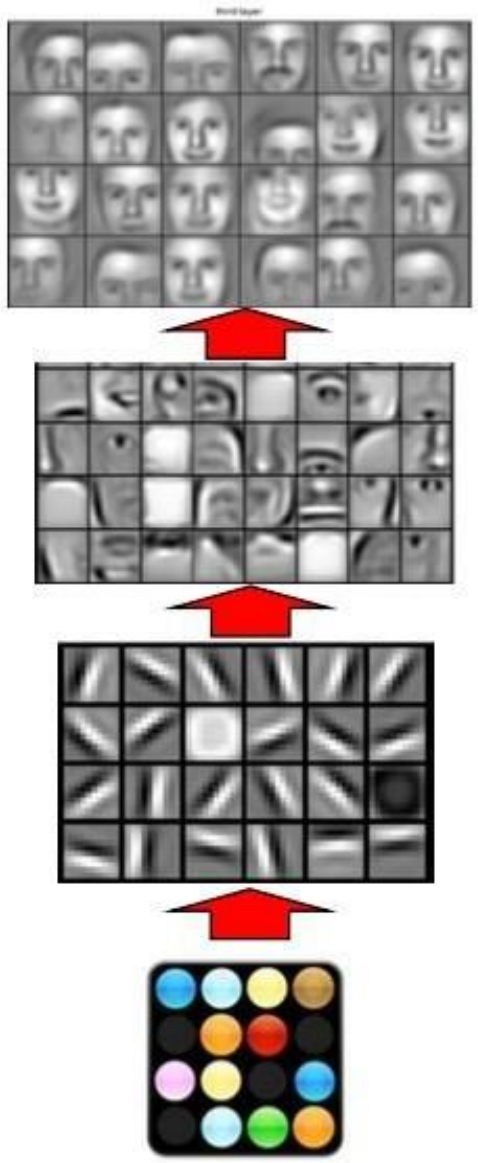


Textons



GLOH

Deep Learning: learn representations!



So, 1. **what exactly is deep learning** ?

And, 2. **why is it generally better** than other methods on image, speech, and certain other data types?

The short answers

- **Deep Learning' means** using a **neural network** with **several layers of nodes** between input and output

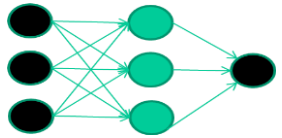
- **The series of layers between input & output feature identification and processing in a series of stages the same as our brains.**

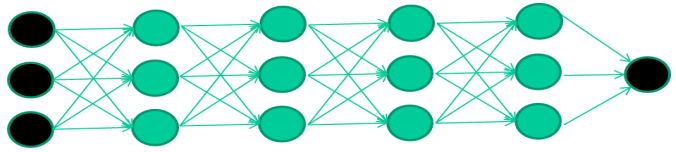
hmmm… OK, but:

3. **Multilayer neural networks have been around for 25 years. What's actually new?**

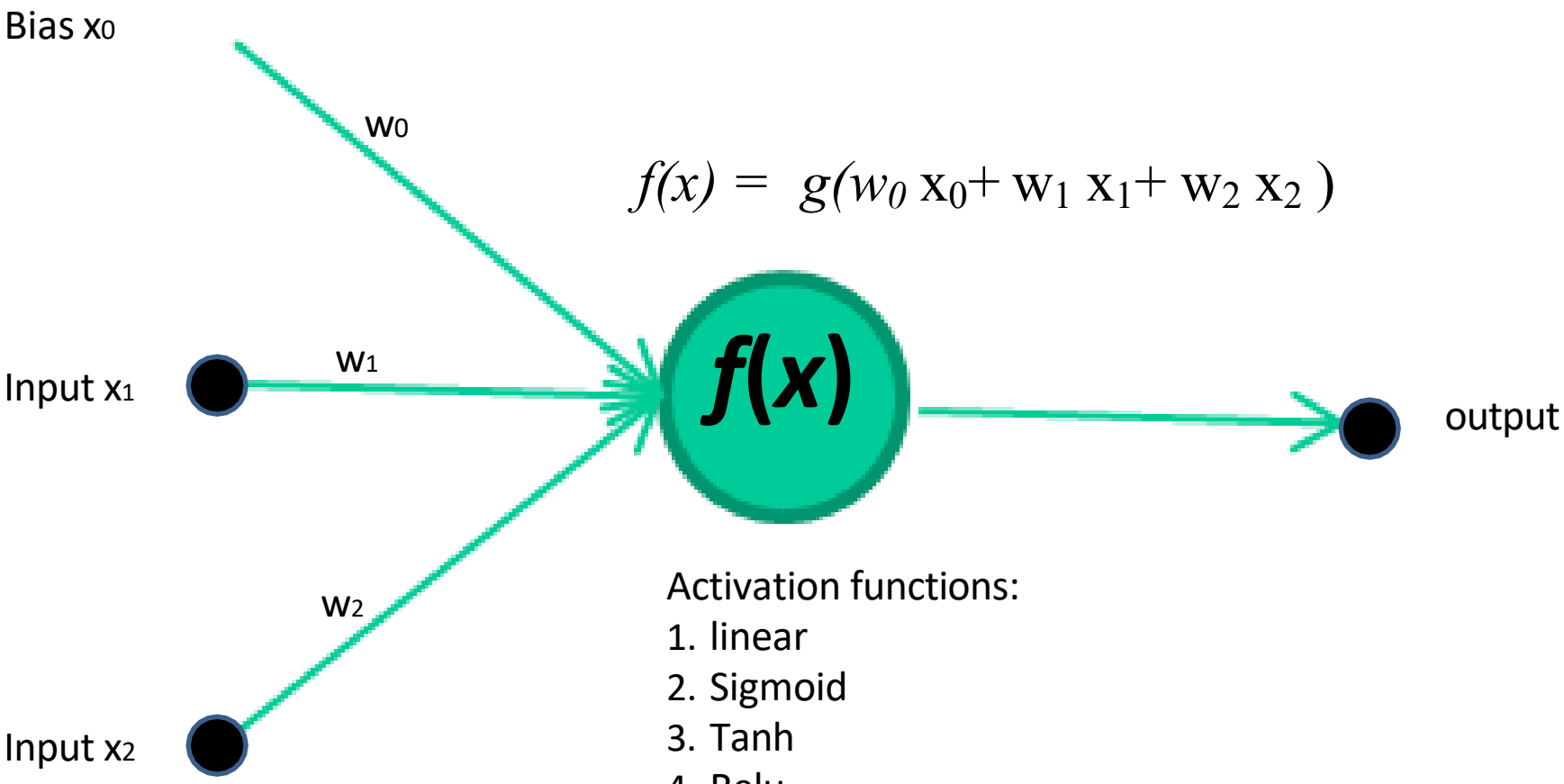
Single layer algorithms are not good at learning the weights for networks with more hidden layers

what's new is: **algorithms for training many–later networks**





Single Unit, Input, weights, activation function, output



Activation functions:

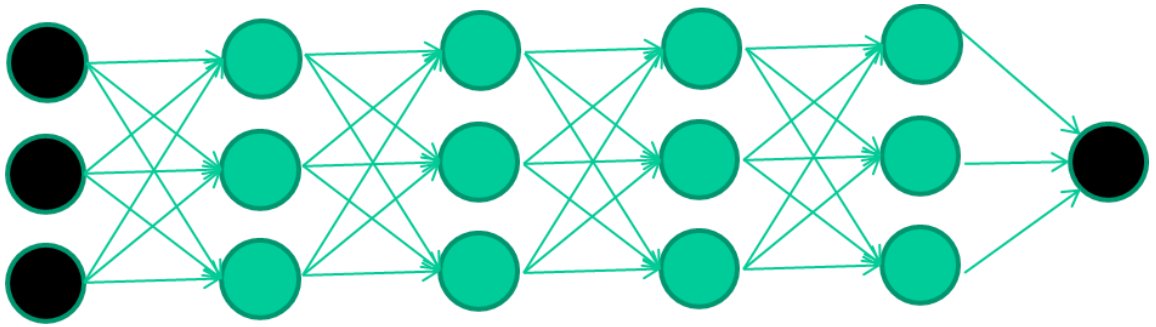
1. linear
2. Sigmoid
3. Tanh
4. Relu
5. Softmax

etc.

A dataset

Fields			***class***
1.4	2.7	1.9	0
3.8	3.4	3.2	0
6.4	2.8	1.7	1
4.1	0.1	0.2	0
etc …			

Train the deep neural network

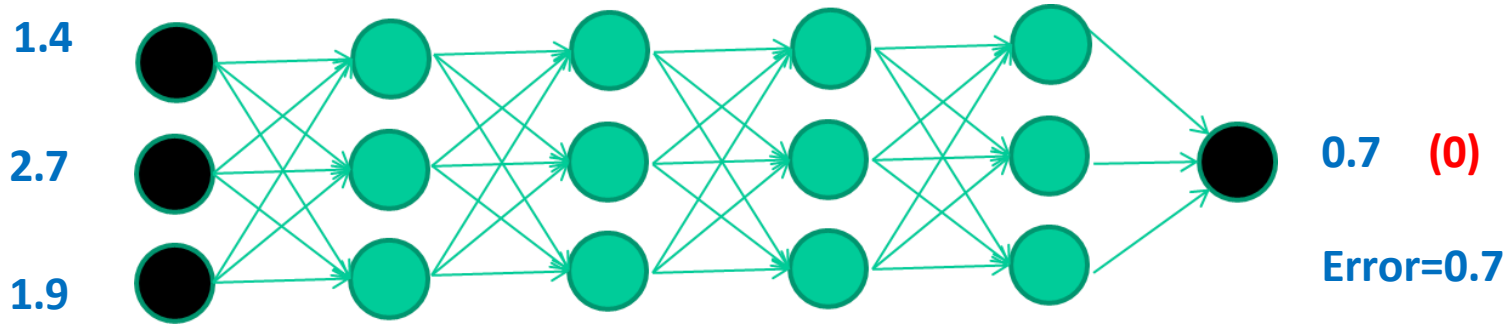


A dataset

Fields			***class***
1.4	2.7	1.9	0
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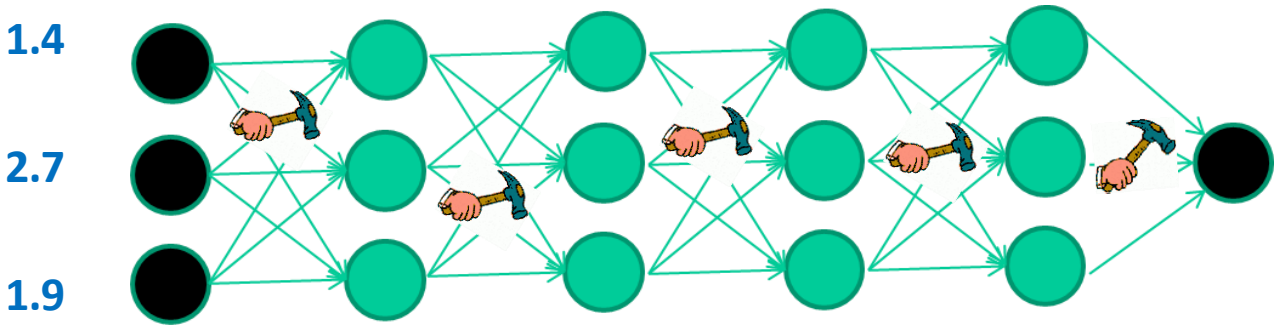
etc …

Initialize with random weights



Compare with the target output

Adjust weights based on error



Repeat this thousands, maybe millions of times – each time taking a random training instance, and making slight weight adjustments

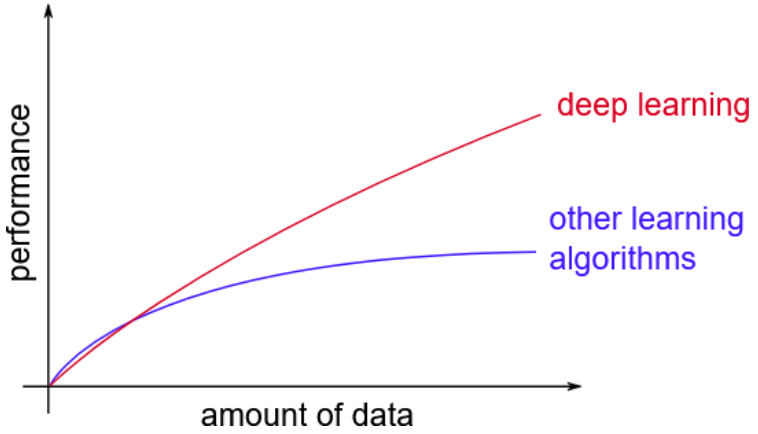
Algorithms for weight adjustment are designed to make changes that will reduce the error

Why Deep Learning Now

- **Better algorithms and Understanding**
- **Computing Power(GPUs, TPUs)**
- **Data with labels**
- **OneSource tools and models**









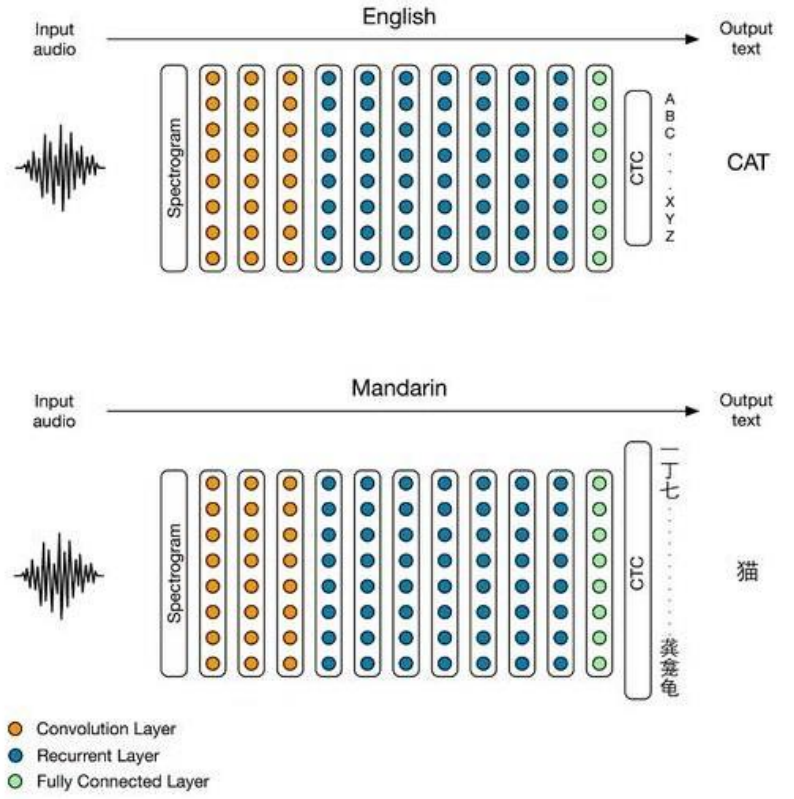
DL Today: Speech to Text











[Baidu 2014]

DL Today: Vision



[Krizhevsky 2012]



[Ciresan et al. 2013]



[Faster R-CNN - Ren 2015]

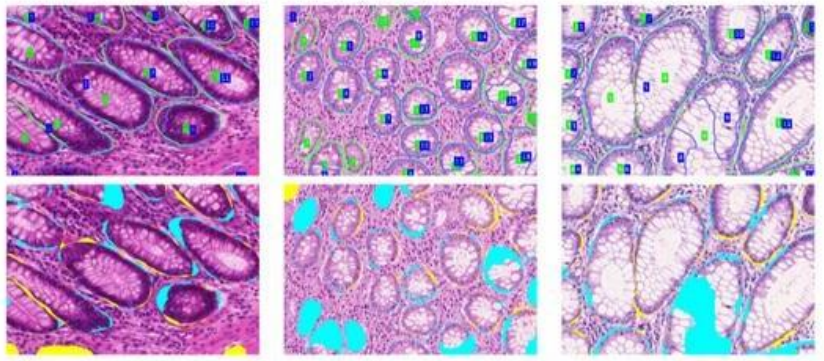


[NVIDIA dev blog]

DL Today: Vision



[Stanford 2017]



[Nvidia Dev Blog 2017]

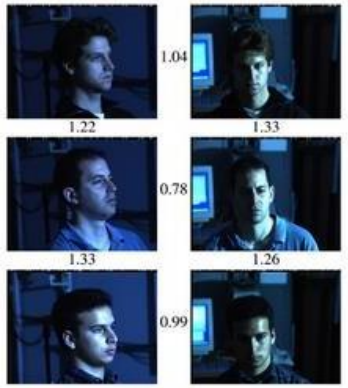
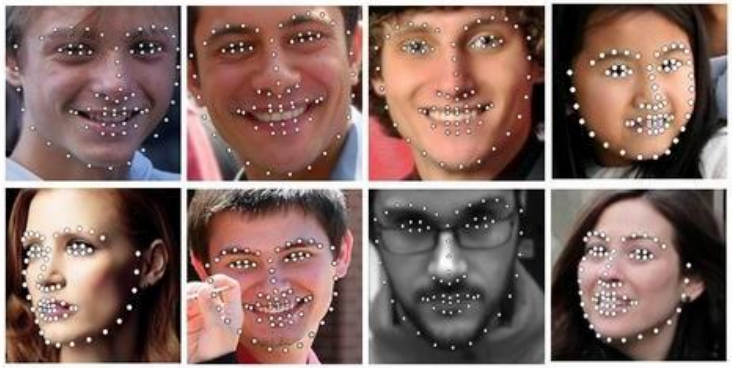


Figure 1. **Illumination and Pose invariance.**

[FaceNet - Google 2015]



[Facial landmark detection CUHK 2014]

DL Today: NLP



[Google Inbox Smart Reply]



[Amazon Echo / Alexa]

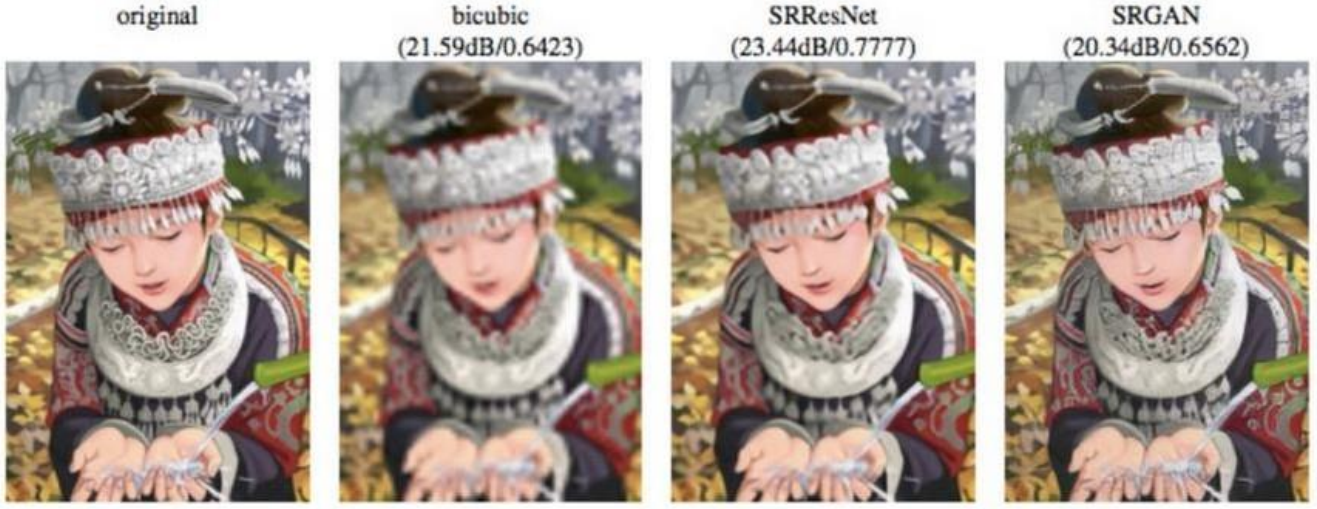
DL Today: Image Translation



[DeepDream 2015]



[Gatys 2015]



[Ledig 2016]

DL Today: Generative Models



Sampled celebrities [Nvidia 2017]



StackGAN v2 [Zhang 2017]